[Total No. of Question -04] [Total No. of Pages - 02]

G H Raisoni International Skill Tech University, Pune

S.Y B. Tech (TERM-IV)

CAE I- Summer 2026 (2024 Pattern)

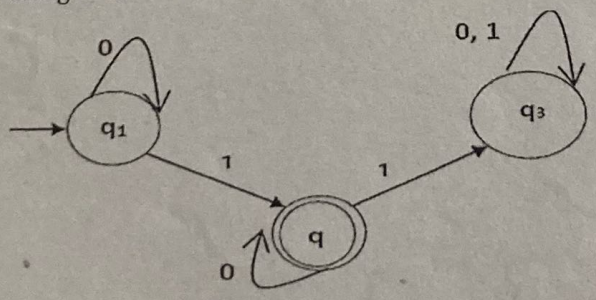
Theory of Computation (25BTPCL404)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	Construct DFA which accepts all strings which ends with abb over ∑={a, b}. Construct Transition Diagram, Transition Table and show acceptance of one Valid string.	[5]	1	3
b)	Define the following terms with example. 1. DFA 2. NFA 3. Transition Diagram 4. Acceptance of string 5. Language	[5]	1	2
	OR			
Q 2.(a)	Design an DFA over the alphabet Σ = {a, b} that accepts only the strings that start with "ab". Construct Transition Diagram, Transition Table and show acceptance of one Valid string.	[5]	1	3
(b)	Differentiate between NFA and DFA. Give five points.	[5]	1	2
Q 3.(a)	Find regular expression for 1. The strings that ends with "abb" over Σ = {a, b} 2. Strings containing the substring "aa" over Σ = {a, b}	[5]	2	2
(b)	Construct a regular expression corresponding to the automata given below using Ardens Theorem.	[5]	2	3



OR